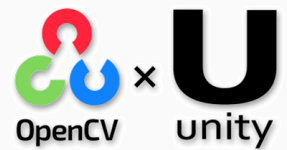
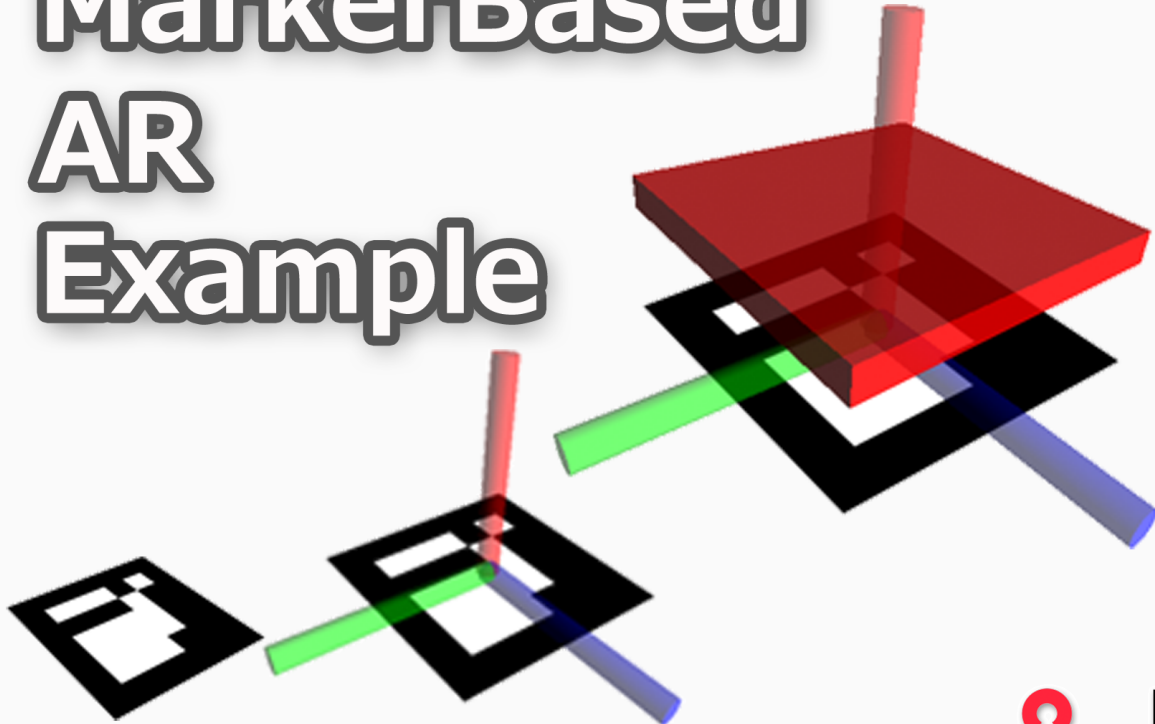


MarkerBased AR Example 1.3.0

MarkerBased AR Example



About MarkerBased AR Example

MarkerBased AR Example is an **example project** demonstrating **Marker-Based Augmented Reality** using [OpenCV for Unity](#). This project showcases how to implement **AR with printed markers** by detecting and recognizing NxN block markers and displaying **3D models** in `WebCamTexture` in real-time.

The execution of this asset is required [OpenCV for Unity](#).



Our Asset Features



Cross Platform Support

- **Mobile:** iOS & Android
- **Desktop:** Windows & macOS & Linux Standalone support
- **Web:** WebGL support
- **ChromeOS:** ChromeOS support
- **XR:** visionOS support (beta) & Windows10 UWP support
- **Editor:** Support for preview in the Editor

Marker-Based AR Technology

- **NxN Block Marker Detection:** Detects and recognizes NxN block markers in real-time
- **Real-time Processing:** Displays 3D models in `WebCamTexture` in real-time
- **Multiple Tracking Methods:** Supports various tracking approaches including gyro sensor integration

Technical Implementation

- **Code Rewrite:** This project's code is a rewrite of [MasteringOpenCV Chapter2_iPhoneAR](#) using [OpenCV for Unity](#)
- **Real-time Processing:** Optimized for real-time marker-based AR performance
- **Easy Integration:** Designed to work seamlessly with Unity's development environment

Example Scenes

- **Texture2DMakerBasedARExample** - By detecting the marker from `Texture2D`, display AR model
- **MultiSourceMakerBasedARExample** - By detecting a marker from `WebCamTexture` or video file, display AR model in real-time
- **GyroSensorMakerBasedARExample** - By detecting a marker from `WebCamTexture` or video file, display AR model in real-time with gyro sensor

Quick Links

- **Official Site** - Main product page
 - **Example Code** - GitHub repository with examples
 - **Android Demo** - Downloadable Android demo
 - **WebGL Demo** - Online WebGL demo
 - **Demo Video** - YouTube demonstration video
-

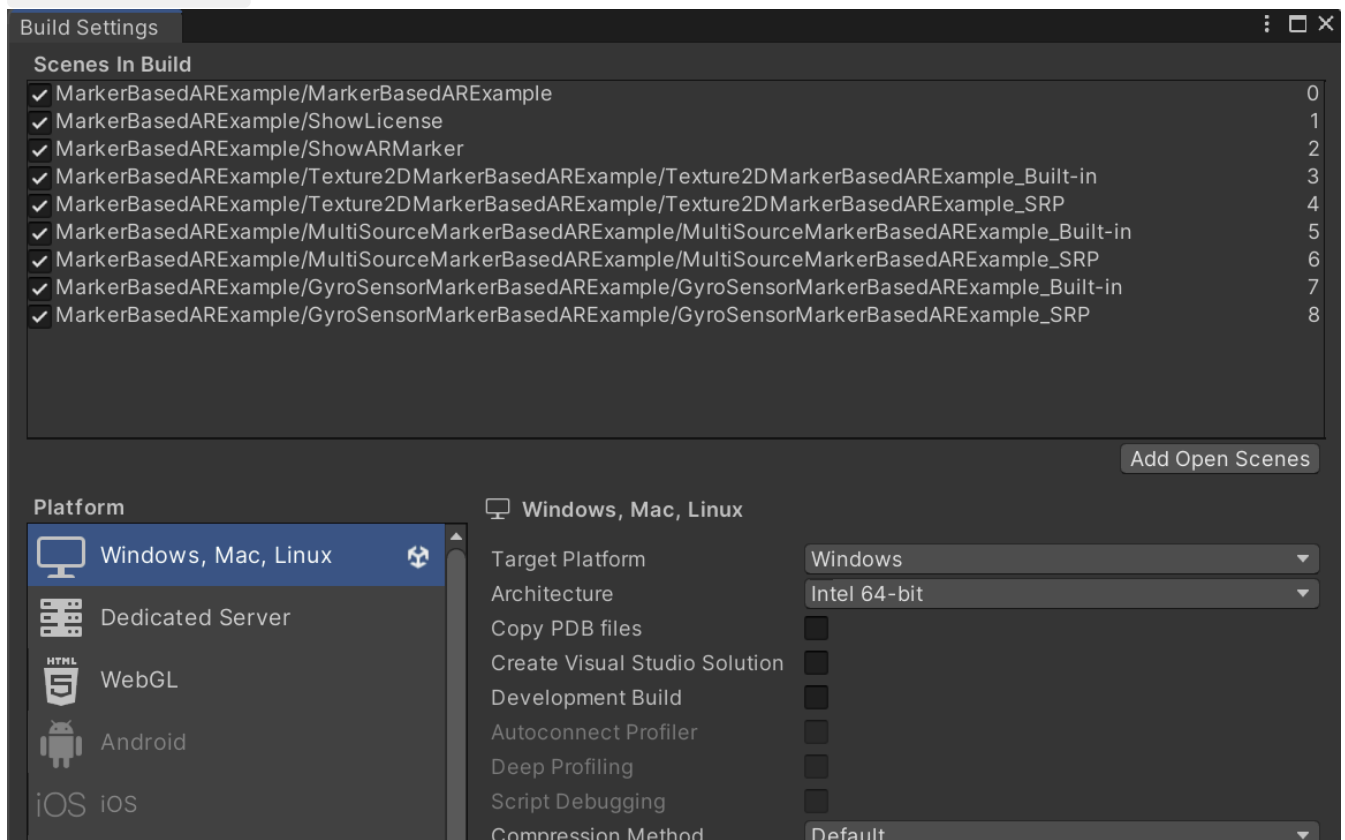
System Requirements

Platform	Requirement
Windows Standalone & Editor	Windows 8 or later
Mac Standalone & Editor	macOS 10.13 or later
Linux Standalone & Editor	Ubuntu 18.04 or later
Android	API level 24 or later
iOS	iOS Version 12.0 or later
visionOS (beta)	visionOS 1 or later

Setup Guide for Getting Started

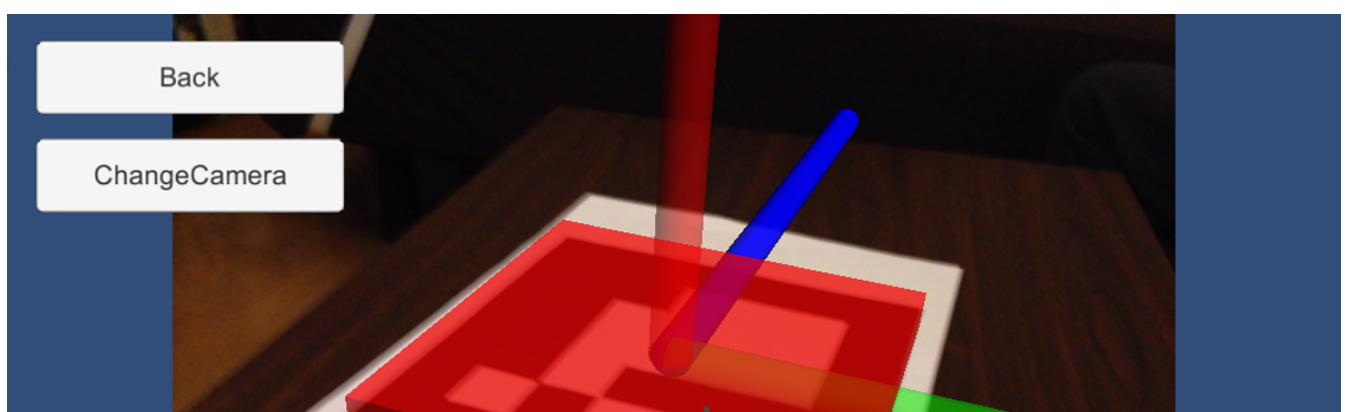
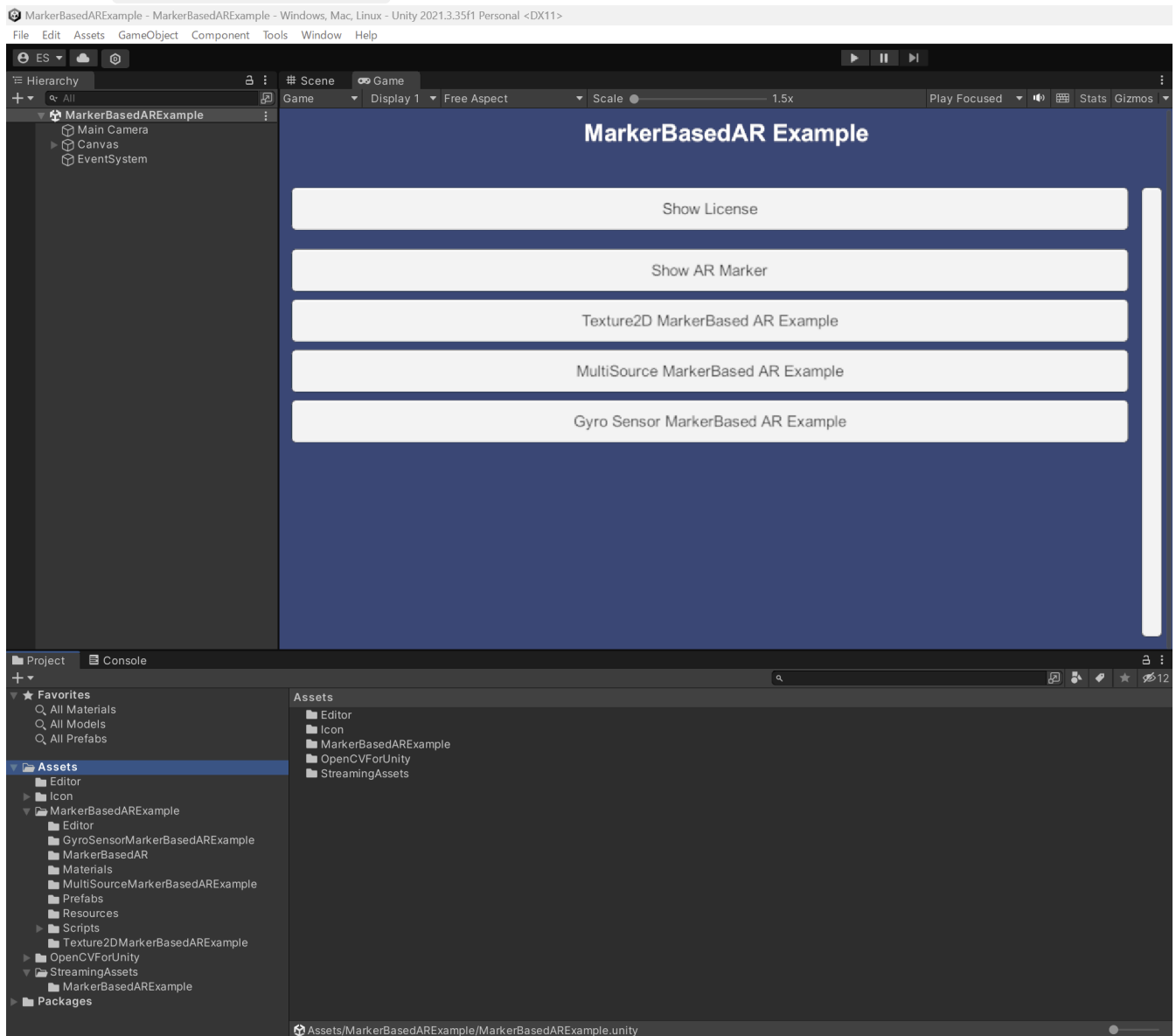
Quick Setup Procedure

1. Import [OpenCV for Unity](#) from the `Package Manager` .
2. Import [MarkerBased AR Example](#) from the `Package Manager` .
3. Add all of the `***.unity` files in the `MarkerBasedARExample` folder to `Build Settings` → `Scenes In Build` .





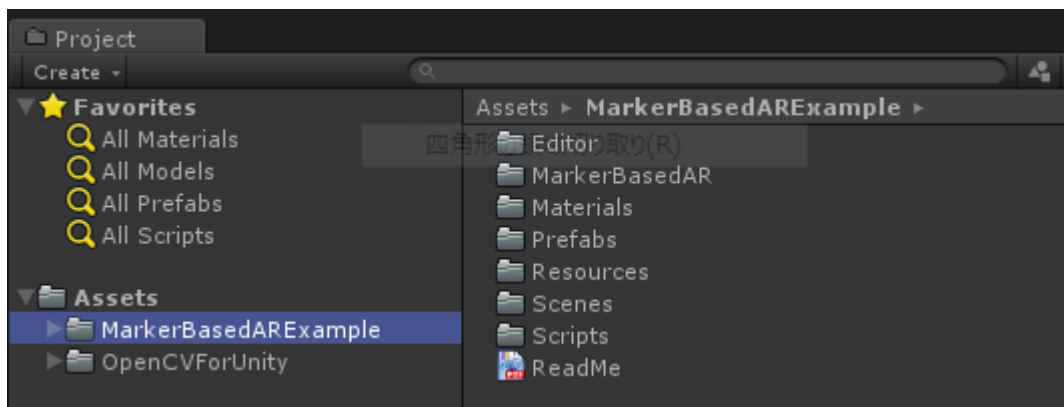
4. Run the `MarkerBasedARExample` scene.





5. Print the marker image (`MarkerBasedARExample/Resources/maker.png`).

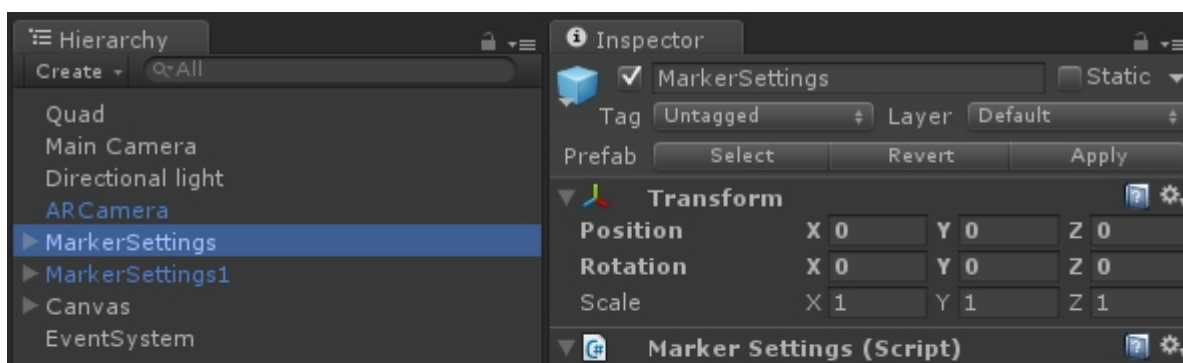
Screenshot after the Setup

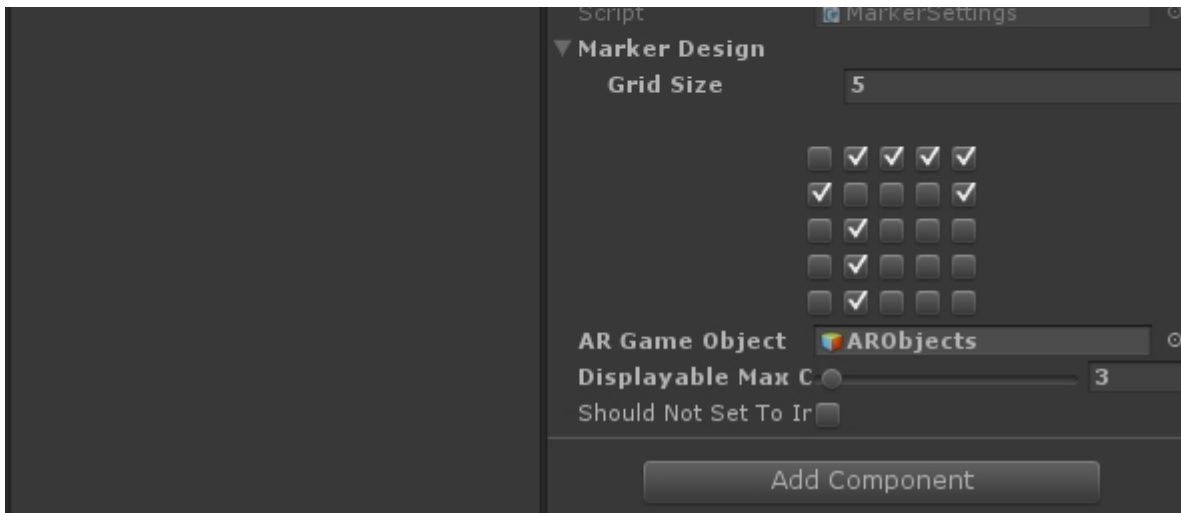


? Q & A

Q1. I want to change the Marker Image.

A1. Marker design is changeable from the Inspector.





Version Changes

- 1.3.0** [Common] Updated for OpenCV for Unity v3.0.4. (This asset requires OpenCVforUnity 3.0.4 or later.)
- 1.2.9** [Common] Updated for OpenCV for Unity v3.0.0. (This asset requires OpenCVforUnity 3.0.0 or later.)
- 1.2.8** [Common] Updated for OpenCV for Unity v2.6.4. (This asset requires OpenCVforUnity 2.6.4 or later.)
- 1.2.7** [Common] Changed the minimum supported version to Unity2021.3.35f1. [Common] Separated the examples using the Built-in Render Pipeline and Scriptable Render Pipeline.
- 1.2.6** [Common] Updated for OpenCV for Unity v2.5.9. (This asset requires OpenCVforUnity 2.5.9 or later.)
- 1.2.5** [Common] Updated for OpenCV for Unity v2.5.0. (This asset requires OpenCVforUnity 2.5.0 or later.)
- 1.2.4** [Common] Updated for OpenCV for Unity v2.4.2. (This asset requires OpenCVforUnity 2.4.2 or later.) [Common] Refactored the script.
- 1.2.3** [Common] Updated for OpenCV for Unity v2.3.8. (This asset requires OpenCVforUnity 2.3.8 or later.)
- 1.2.2** [Common] Updated for OpenCV for Unity v2.3.3. (This asset requires OpenCVforUnity 2.3.3 or later.)
- 1.2.1** [Common] Fixed getMarkerId() method. [Common] Updated WebCamTextureToMatHelper.cs v1.0.1.
- 1.2.0** [Common] Changed the name of asset project. ("Sample" to "Example")
- 1.1.9** [Common] Fixed WebCamTextureToMatHelper.cs. (flipVertical and flipHorizontal flag)
- 1.1.8** [WebGL] Added WebGL(beta) support. (Unity5.3 or later)
- 1.1.7** [Common] Changed namespace to OpenCVMarkerBasedAR. (To avoid namespace and classname conflict.) [Common] Fixed CS0618 warnings:

`UnityEngine.Application.LoadLevel(string)` is obsolete: Use `SceneManager.LoadScene`.

1.1.6 [Common] Added namespace. [Common] Added `flipVertical` flag, `flipHorizontal` flag and `GetWebCamDevice()` method to `WebCamTextureToMatHelper.cs`.

1.1.5 [Common] Changed to methods of moving the AR object.

1.1.4 [Common] Support for "OpenCV for Unity 2.0.0".

1.1.3 [Common] Fixed `WebCamTextureToMatHelper.cs`. (Add `didUpdateThisFrame ()` method)

1.1.2 [Common] Renewed the samples using `WebCamTextureToMatHelper`. (Supports all screen orientation.)

1.1.1 [Common] Added Marker Design Inspector. (Improved procedures to change the marker image)

1.1.0 [Common] Change to use `uGUI` in `SampleScene`.

1.0.9 [iOS] Fix `WebCamTexture` bug of `SampleScene` in Unity5.2.

1.0.8 [Common] Rewrite `SampleScene`.

1.0.7 [Common] Add the code to support Beta Version of "OpenCV for Untiy" based on "OpenCV3.0.0".

1.0.6 [Common] Fix `SampleScene`.

1.0.5 [Common] Fix `SampleScene`. [Common] Change Property of Platform Dependent Compilation from `UNITY_IPHONE` to `UNITY_IOS`.

1.0.4 [Common] Fix the direction of rotation of the mat that is converted from `WebCamTexture`.

1.0.3 [Common] Fix direction of `WebCamTexture`.

1.0.2 [Common] Divide asset for Unity4 and Unity5.

1.0.1 [Common] Support for "OpenCV for Unity 1.0.9"

1.0.0 Initial version